MATHEMATICS

Applications and interpretation

Higher Level | Set C | Paper 2

Challenge mock examination

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A graphic display calculator and a clean Mathematics: applications and interpretation formula booklet are required.

• Answer all questions. Show working and explain mathematical reasoning in context.

• All questions are compulsory. There is no optional section.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures, unless a question specifies a different accuracy.

• Keep unrounded values in subsequent calculations. State units where relevant.

• Diagrams are not necessarily drawn to scale. Use the stated data and conditions.

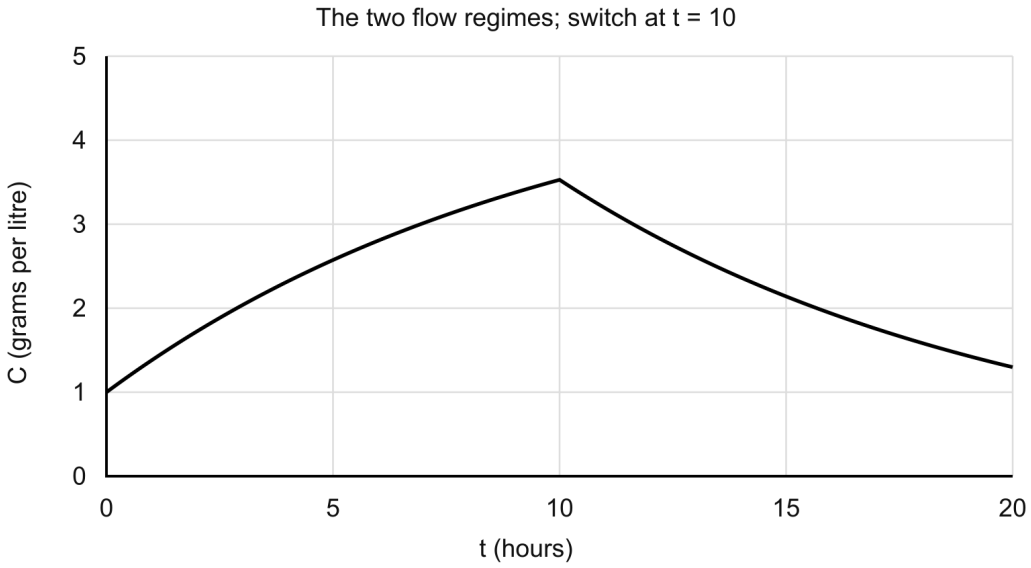
• The number in square brackets gives the maximum mark for a part.

Thriving Scholars original practice | 2026 challenge series

Question 1 | Question paper

Question 1 [Maximum mark: 18]

A well-mixed tank contains 1000 litres of water. Liquid flows in and out at 100 litres per hour, keeping the volume fixed. Initially the pollutant concentration is 1 gram per litre. For the first ten hours the incoming concentration is 5 grams per litre, so the tank concentration $C(t)$ satisfies $C\prime = 0.5 - 0.1C$. At $t = 10$ the inflow is changed to clean water; the outflow rate remains unchanged. Assume instantaneous mixing and no other pollutant loss.



(a) Solve the differential equation for $0 \leq t \leq 10$ and find $C(10)$. [4]

(b) Use Euler's method with step size 2.5 hours to estimate $C(10)$ from the original differential equation. State whether it overestimates or underestimates the exact value, with a reason. [3]

(c) Write the differential equation for $t > 10$ and find its solution, preserving concentration continuity at the switch. [3]

(d) Find the total time for which the pollutant concentration exceeds 2 grams per litre. [4]

Question 1 | Question paper

(e) Calculate the mass of pollutant leaving through the outlet during the first 20 hours. Verify your result using a mass balance. [4]

Question 2 | Question paper

Question 2 [Maximum mark: 18]

A courier must start at A, visit each of B, C, D, E and return to A. The symmetric table gives the shortest available travel distances, in kilometres, between every pair. A tour is assessed on this complete distance graph; travelling along a shortest connection may pass through another site without counting it as its scheduled visit.

	A	B	C	D	E
A	0	7	9	8	12
B	7	0	6	10	9
C	9	6	0	5	7
D	8	10	5	0	6
E	12	9	7	6	0

(a) Apply the nearest-neighbour algorithm starting at A. State the tour and its length, explaining why it is an upper bound. [4]

(b) Find a lower bound by deleting A, finding a minimum spanning tree on the remaining sites, and then adding the two smallest connections from A. Explain the lower-bound argument. [5]

(c) Find a tour shorter than the nearest-neighbour tour in (a), and use it with (b) to improve the interval containing the optimal length. [3]

(d) The direct path CD is closed. All the other listed direct paths remain available. Recalculate shortest travel distances as needed, apply nearest neighbour from A, and find a deleted-A lower bound. Hence determine the new optimal tour length. [5]

(e) Explain why closing the direct path CD does not increase the best known tour length from part (c). [1]

Question 3 | Question paper

Question 3 [Maximum mark: 18]

Two independent random samples compare completion times for procedures A and B. The populations may be treated as normal with a common unknown variance. The reported standard deviations are unbiased sample standard deviations.

Procedure	n	Mean (minutes)	Sample standard deviation (minutes)
A	12	52.4	4.1
B	15	55.0	3.6

Use a pooled two-sample t test when comparing means.

(a) Find the pooled estimate of the common standard deviation and the estimated standard error of the difference between sample means. [3]

(b) At the 5% significance level, test whether procedure A has a smaller population mean completion time than procedure B. State hypotheses, statistic, degrees of freedom and p-value. [5]

(c) Using only the sample from A, find a 95% confidence interval for its population mean completion time. State the distribution and degrees of freedom used. [4]

(d) Both timers are found to have understated every recorded time by exactly 2 minutes. Explain the effects of correcting this error on the test in (b) and the interval in (c). [3]

Question 3 | Question paper

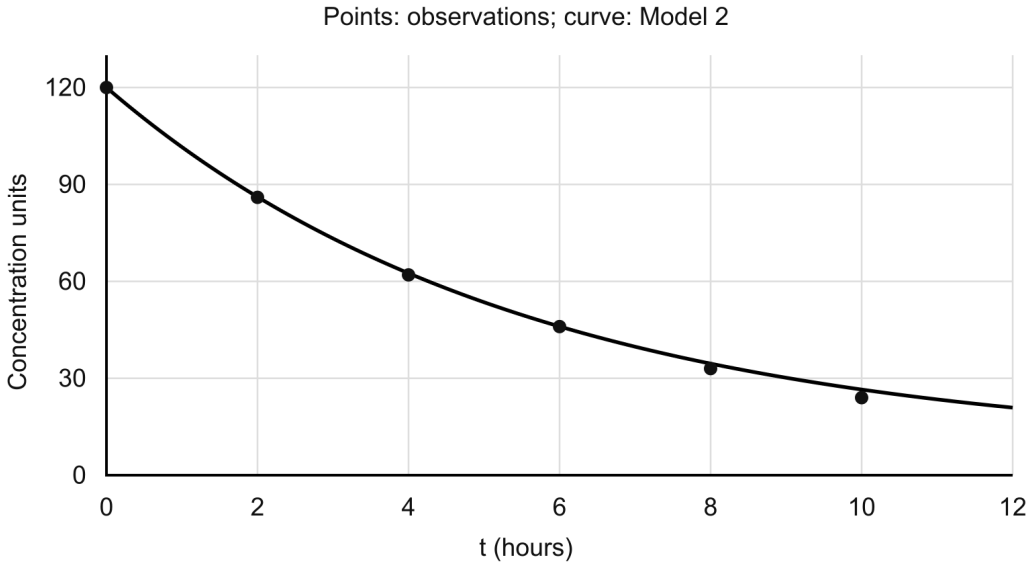
(e) A manager proposes replacing each operator's one measurement by five repeat measurements and treating all repeats as independent observations. Explain why this could produce misleading evidence, and suggest a better design. [3]

Question 4 [Maximum mark: 18]

A sealed chamber contains a gas whose concentration is measured in arbitrary units after ventilation begins. The observations are:

t (hours)	0	2	4	6	8	10
Concentration	120	86	62	46	33	24

Model 1 is $C_1(t) = Ae^{bt}$, fitted by linear regression of $\ln C$ on t. A physical model which allows a background concentration is $C_2(t) = 8 + 112e^{-0.18t}$, for $t \geq 0$.



(a) Find the parameters of Model 1, showing the linearized equation. [4]

(b) Using Model 1, find the first time at which concentration falls below 30. [3]

(c) For each model calculate the sum of squared residuals in the original concentration units. Which fits these six observations better on this criterion? Explain why this comparison does not by itself validate long-term predictions. [4]

Question 4 | Question paper

(d) For Model 2, find the time taken for the excess concentration above the background to halve. Explain why the same time does not generally halve the total concentration. [3]

(e) Using Model 2, calculate the integrated concentration over the first 12 hours and hence the mean concentration over that interval. State appropriate units for both. [4]

Question 5 [Maximum mark: 19]

A robot's position in a plane is represented by the complex number z. Successive commands apply $z_{n+1} = (0.6 + 0.2i)z_n + (2 - i)$, with $z_0 = 1 + 3i$. Coordinates are in metres. The linear part of this affine map acts before the translation.

(a) Find z_1 and z_2 in Cartesian form. [3]

(b) Write the real 2×2 matrix for the linear part. State its scale factor, rotation angle and area scale factor. [4]

(c) Find the fixed position z_* satisfying $z_* = (0.6 + 0.2i)z_* + (2 - i)$. [3]

(d) By subtracting the fixed-position equation from the recurrence, obtain a formula for $z_n - z_*$. Verify that your formula satisfies both the recurrence and the initial condition. [4]

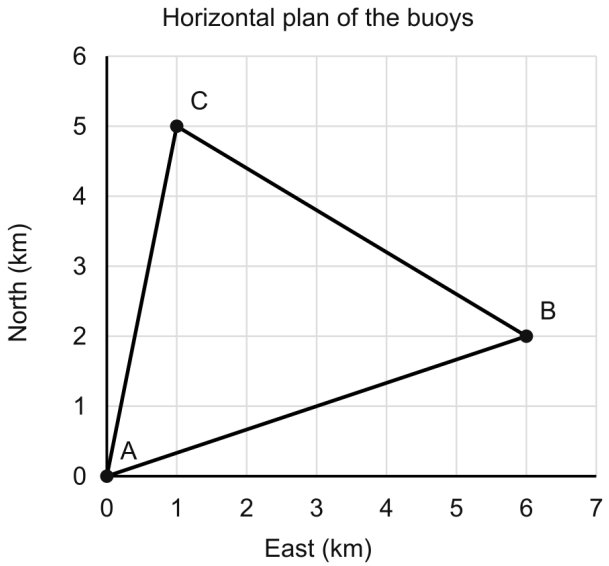
(e) Find the smallest integer n for which the robot is within 0.05 metres of the fixed position. [3]

(f) Explain why the positions approach the fixed point in a rotating pattern rather than along a single straight line. [2]

Question 6 | Question paper

Question 6 [Maximum mark: 19]

In a coastal survey, east, north and vertical coordinates are measured in kilometres. Buoys are at $A(0, 0, 0)$, $B(6, 2, 0)$ and $C(1, 5, 0)$; a drone is at $D(3, 2, 4)$. Bearings are measured clockwise from north. In parts (c) and (d), a uniform horizontal current has velocity $(0.2, 0.3)$ km per hour in east-north coordinates.



(a) Find the area of the triangle formed by the three buoys. [3]

(b) Find the angle between the sight lines AB and AD. [3]

(c) A boat has velocity $(2, 0.5)$ km per hour relative to the water. Find its bearing relative to the fixed shore. [3]

(d) A second boat leaves A and travels directly to C at a constant speed of 2.5 km per hour relative to the water. Find the travel time and the heading it must maintain relative to the water. Show the equation used to allow for the current. [6]

Question 6 | Question paper

(e) A straight floating cable is stretched along the segment AB. Find the point on the cable closest to C and the minimum distance. Check that your point lies on the cable segment. [4]

End of question paper

MATHEMATICS

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Worked solution book

Complete this paper before consulting the worked solutions. Each part has method (M), accuracy (A) and reasoning (R) points. The number following a code is its mark value. Award only the points supported by the candidate's retained working.

Accept mathematically equivalent valid methods. Numerical answers may use technology, but should be supported by equations, test definitions, bounds, labelled graphs or other appropriate mathematical working. Retain unrounded intermediate values.

Use the portal to enter a whole-number self-assessed mark for each part. These are independent practice papers; the self-assessed report does not calculate an official IB grade.

Question 1 | Worked solutions

Question 1 [Maximum mark: 18]

(a) [4 marks]

M1 Separate variables or recognize $C = 5 + Ae^{-0.1t}$.

A1 The initial condition gives $A = -4$.

A1 $C(t) = 5 - 4e^{-0.1t}$ for $0 \le t \le 10$.

A1 $C(10) = 3.528482$ grams per litre.

(b) [3 marks]

M1 Use $C_{n+1} = C_n + 2.5(0.5 - 0.1C_n)$, starting at $C_0 = 1$.

A1 The values are $1, 2, 2.75, 3.3125, 3.734375$, so the estimate is 3.734375.

R1 This is an overestimate; the exact curve is increasing and concave down, so forward tangent steps lie above it.

(c) [3 marks]

A1 With clean inflow, $C\prime = -0.1C$.

M1 Use the exact value $C(10)$ as the new initial condition.

A1 $C(t) = 3.528482e^{-0.1(t-10)}$ for $t \ge 10$, using the unrounded coefficient.

(d) [4 marks]

M1 Solve $5 - 4e^{-0.1t} = 2$ in the first regime and $C(10)e^{-0.1(t-10)} = 2$ in the second.

A1 The upward crossing is $t = 2.876821$ h and the downward crossing is $t = 15.67721$ h.

R1 The concentration exceeds 2 only between these crossings.

A1 The total duration is 12.80039 hours.

(e) [4 marks]

M1 Outflow mass is $100\{\int_0^{10} C(t)\,dt + \int_{10}^{20} C(t)\,dt\}$ grams, using the appropriate expressions.

A1 The removed mass is 4701.944 grams.

M1 Initially there are 1000 g and the first ten hours add $100 \times 5 \times 10 = 5000$ g; the final mass is $1000C(20)$.

A1 $1000 + 5000 - 1000C(20) = 4701.944$ grams, agreeing with the integral.

Question 2 [Maximum mark: 18]

(a) [4 marks]

M1 At each step choose the nearest unvisited site.

A1 The tour is $A - B - C - D - E - A$.

A1 Its length is $7 + 6 + 5 + 6 + 12 = 36$ km.

R1 It is a feasible tour, so the optimal tour cannot be longer than this particular tour.

(b) [5 marks]

M1 On B, C, D, E, choose $CD = 5$, $BC = 6$ and $DE = 6$.

A1 The MST weight is 17.

A1 The two smallest edges from A are $AB = 7$ and $AD = 8$.

A1 The lower bound is $17 + 7 + 8 = 32$ km.

R1 Deleting A from any tour leaves a spanning path, whose weight is at least the MST weight, and the two removed connections have sum at least $7 + 8$.

(c) [3 marks]

M1 Consider the feasible tour $A - B - C - E - D - A$.

A1 Its length is $7 + 6 + 7 + 6 + 8 = 34$ km.

A1 Thus $32 \leq L_{\text{opt}} \leq 34$ km; these bounds alone do not prove the tour is optimal.

(d) [5 marks]

M1 The new shortest $C - D$ connection is $C - E - D$ of length 13; all other table entries remain unchanged.

A1 Nearest neighbour gives $A - B - C - E - D - A$, length 34.

M1 The deleted-A MST now uses $BC = 6$, $DE = 6$, $CE = 7$, weight 19.

A1 The new lower bound is $19 + 7 + 8 = 34$.

R1 The matching upper and lower bounds prove the optimum is 34 km.

(e) [1 marks]

R1 The 34-km feasible tour from (c) does not use CD and remains available with the same length. This observation does not by itself establish the original optimal length.

Question 3 | Worked solutions

Question 3 [Maximum mark: 18]

(a) [3 marks]

M1 $s_p^2 = \{11(4.1)^2 + 14(3.6)^2\}/25$.

A1 $s_p = 3.828054$ minutes.

A1 The standard error is $s_p\sqrt{1/12 + 1/15} = 1.482599$ minutes.

(b) [5 marks]

A1 $H_0 : \mu_A = \mu_B$, $H_1 : \mu_A < \mu_B$.

M1 $t = (52.4 - 55.0)/(s_p\sqrt{1/12 + 1/15})$.

A1 $t = -1.753677$ with $12 + 15 - 2 = 25$ degrees of freedom.

A1 The lower-tail p-value is 0.04586992.

R1 Since this is less than 0.05, reject H_0; the data support a smaller mean for A under the stated assumptions.

(c) [4 marks]

M1 Use $52.4 \pm t_{0.975,11}(4.1/\sqrt{12})$.

A1 This uses the t distribution with 11 degrees of freedom, since the population variance is unknown.

A1 The critical value is 2.200985.

A1 The confidence interval is $(49.79498, 55.00502)$ minutes.

(d) [3 marks]

R1 Both sample means increase by 2, leaving their difference, both sample standard deviations and the pooled statistic unchanged.

R1 The p-value and conclusion in (b) are therefore unchanged.

A1 The interval for A shifts upward by 2, to $(51.79498, 57.00502)$ minutes.

(e) [3 marks]

R1 Repeats on the same operator may be correlated and do not supply five independent operators.

R1 Treating them as independent can underestimate standard error and exaggerate significance.

R1 Sample more independent operators, or summarize each operator's repeats before an analysis that respects the sampling unit.

Question 4 | Worked solutions

Question 4 [Maximum mark: 18]

(a) [4 marks]

M1 Fit $\ln C = \ln A + bt$.

A1 The fitted intercept is 4.780068 and slope is -0.1602743.

A1 $A = e^{4.780068} = 119.1124$.

A1 Thus $C_1(t) = 119.1124e^{-0.1602743t}$, with unrounded regression coefficients retained for subsequent work.

(b) [3 marks]

M1 Solve $Ae^{bt} = 30$.

A1 The crossing is $t = \ln(30/A)/b = 8.603189$ hours.

R1 Since $b < 0$, concentration is below 30 after this time.

(c) [4 marks]

M1 Calculate $\sum_i (C_i - C_j(t_i))^2$ for each $j = 1, 2$ on the original scale.

A1 The sums are 1.752691 for Model 1 and 8.963826 for Model 2.

R1 Model 1 has the smaller original-scale sum and fits these observations better by this criterion.

R1 These are short-term observations; a physical background and future ventilation conditions can matter beyond the observed interval.

(d) [3 marks]

M1 The excess is $C_2 - 8 = 112e^{-0.18t}$, so solve $e^{-0.18h} = 1/2$.

A1 $h = \ln 2/0.18 = 3.850818$ hours.

R1 The constant background 8 does not decay, so halving the excess does not halve the sum of excess and background.

(e) [4 marks]

M1 Use $\int_0^{12} (8 + 112e^{-0.18t})\,dt$.

A1 The integral is 646.4644 concentration-unit hours.

M1 Divide the integral by the interval length, 12 hours.

A1 The mean concentration is 53.87203 concentration units.

Question 5 | Worked solutions

Question 5 [Maximum mark: 19]

(a) [3 marks]

M1 Substitute z_0 into the recurrence and multiply the complex numbers.

A1 $z_1 = 2 + i$.

A1 $z_2 = 3$.

(b) [4 marks]

A1 The matrix is

$$\begin{pmatrix} 0.6 & -0.2 \\ 0.2 & 0.6 \end{pmatrix}$$

A1 The length scale factor is $\sqrt{0.4} = 0.6324555$.

A1 The rotation is 18.43495° anticlockwise.

A1 The area scale factor is the determinant, 0.4; translation does not affect lengths or areas.

(c) [3 marks]

M1 Rearrange to $(0.4 - 0.2i)z_* = 2 - i$.

M1 Divide by $0.4 - 0.2i$, using the conjugate or technology.

A1 $z_* = 5$.

(d) [4 marks]

M1 Let $w_n = z_n - 5$. Then $w_{n+1} = (0.6 + 0.2i)w_n$.

A1 Thus $z_n - 5 = (0.6 + 0.2i)^n(-4 + 3i)$.

R1 At $n = 0$ this gives $z_0 - 5 = -4 + 3i$, agreeing with the initial position.

R1 Multiplying the formula by $0.6 + 0.2i$ gives exactly the next-step recurrence.

(e) [3 marks]

M1 $|z_n - 5| = 5(\sqrt{0.4})^n < 0.05$.

A1 The distances at $n = 11$ and the preceding step are 0.03238172 and 0.0512 metres.

A1 The smallest integer is $n = 11$.

Question 5 | Worked solutions

(f) [2 marks]

R1 At each step the displacement from 5 is rotated by the nonzero angle $18.43495°$.

R1 Its length is multiplied by $0.6324555 < 1$, so it contracts while its direction changes.

Question 6 | Worked solutions

Question 6 [Maximum mark: 19]

(a) [3 marks]

M1 Use $\frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$.

A1 The cross product is $(0, 0, 28)^T$.

A1 The area is $14\ \text{km}^2$.

(b) [3 marks]

M1 Use the dot-product angle formula for $(6, 2, 0)$ and $(3, 2, 4)$.

A1 The dot product is 22 and the product of lengths is $\sqrt{40}\sqrt{29}$.

A1 The angle is $49.76364°$.

(c) [3 marks]

M1 Add the current to obtain shore velocity $(2.2, 0.8)$.

M1 For an east-north vector in this quadrant, bearing is $\tan^{-1}(2.2/0.8)$.

A1 The bearing is $70.01689°$.

(d) [6 marks]

M1 If the travel time is $t > 0$, shore velocity is $(1/t, 5/t)$.

M1 The water-relative velocity is $(1/t - 0.2, 5/t - 0.3)$.

A1 Its magnitude condition is $(1 - 0.2t)^2 + (5 - 0.3t)^2 = 6.25t^2$.

A1 The positive solution is $t = 1.802012$ hours.

A1 The water-relative components are $(0.3549352, 2.474676)$ km per hour.

A1 The required heading is $8.162094°$ clockwise from north.

(e) [4 marks]

M1 Points on the cable are $\mathbf{r} = s(6, 2, 0)$, $0 \le s \le 1$; minimize $|\mathbf{r} - C|^2$ or use a perpendicular dot product.

A1 $s = (C \cdot B)/(B \cdot B) = 16/40 = 0.4$, which is in $[0, 1]$.

A1 The closest point is $(2.4, 0.8, 0)$.

A1 The minimum distance is 4.427189 km.